Joshua Zatz

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Education

Aug ‘20 – May ‘25

University of Illinois at Chicago | Undergraduate in Mechanical Engineering, Minor in Computer Science

- Activities: Society of Automotive Engineers (Active Member); American Society of Mechanical Engineers (Public Outreach Officer)
- 3.8/4.0 GPA cumulative

Professional Experience

Abott Laboratories | Contract Job **Oct’24 – May ‘25**

- Developed a simulation to model electrolyte movement through a porous membrane and trigger a sharp-point RFID/NFC label chip through resistance variation.
- Conducted CFD to determine impact of electrolyte viscosity, flow rate, and concentration on electrical resistivity and triggering mechanism of a smart label.

University of Illinois at Chicago | Undergraduate Research **May ‘24 – May ‘25**

- Provide new ergonomic designs in an impedance controlled assistive exo-skeleton.
- Conduct human trials to evaluate the efficiency of the impedance controller in providing motor-assisted support.
- Test various sensors (FSR, IMU etc.) that act as a backup in case the impedance controller fails on the user.

United States Gypsum | Thermal Engineering Intern **May ‘23 – August ‘23**

- Collaborated in analysis, design and implementation of a heat exchanger system that led to significant cost and carbon emission reduction.
- Measure and test moisture in the mills, effectively reducing gypsum calcination time by ~ 5%.
- Performed Factory Acceptance Tests (FATs) on pipes that deliver natural gas and air to burners.

Projects

FermiLab Computer-Vision Based Robot Arm **May ‘25**

- Utilized Machine Learning (ML) and TensorFlow into a 6-axis manipulator for bolt detection and unscrew/screw operations.
- Designed and 3D printed an end effector that can output 60Nm of torque while keeping the manipulator stable.
- Reduced the operation time of unscrewing/screwing bolts on a flange by 2770%, eliminating the need for human intervention in a radioactive environment.

Heat Exchanger Retrofit Project **Jul ‘23**

- Implemented a heat exchanger in replacement of a boiler in the gypsum wallboard process saving roughly 28Mbtu/year.
- Created an Excel tool to monitor moisture content and dry bulb temperature of air in the mills, ensuring consistent drywall quality.

Exoskeleton Ergonomics **Apr ‘24**

- Developed a model that uses Force Sensing Resistors to detect force and feed the information into an admittance controller (Raspberry pi).
- Redesigned the backplates, braces and kill switch to account for comfort, a 28% weight reduction, and achieve a more modular design.

Additional

Technical Skills: Advanced in CAD, Ansys, 3D printing, Python, MATLAB, C++, Thermal Analysis, 3D Printing

Certifications & Training: SolidWorks, Machine Training

Awards: Deans list (2020,2021,2022,2024), Magna Cum Laude